

EXPERIMENT 10

Aim : To learn Dockerfile instructions, build an image for a sample web application using DOCKERFILE

Theory :

A Dockerfile is a text file that contains a set of instructions used to automate the process of building Docker images. It defines the environment, dependencies, and commands required to run an application inside a container. Docker uses Dockerfiles to ensure consistency, portability, and reproducibility of applications across different systems.

A Dockerfile consists of several important instructions. The FROM instruction specifies the base image (e.g., Ubuntu, Node.js), while WORKDIR sets the working directory inside the container. The COPY or ADD instructions are used to transfer application files into the container. The RUN instruction executes commands during the image build process, such as installing dependencies. The CMD or ENTRYPOINT instruction defines the default command to run the application when the container starts. Additionally, EXPOSE is used to specify the port on which the application will run.

By writing a Dockerfile for a sample web application, developers can build a custom image using the docker build command and run it as a container using docker run. This approach simplifies deployment, ensures environment consistency, and allows the application to run seamlessly across development, testing, and production environments. Docker-based workflows significantly improve scalability and efficiency in modern DevOps practices.

```
C:\Users\Lenovo>docker run redis
1:C 24 Feb 2026 07:57:09.266 * o000o000o000o Redis is starting o000o000o000o
1:C 24 Feb 2026 07:57:09.266 * Redis version=7.2.4, bits=64, commit=00000000, modified=0, pid=1, just started
1:C 24 Feb 2026 07:57:09.266 # Warning: no config file specified, using the default config. In order to specify a config
file use redis-server /path/to/redis.conf
1:M 24 Feb 2026 07:57:09.266 * monotonic clock: POSIX clock_gettime
1:M 24 Feb 2026 07:57:09.267 * Running mode=standalone, port=6379.
1:M 24 Feb 2026 07:57:09.268 * Server initialized
1:M 24 Feb 2026 07:57:09.269 * Ready to accept connections tcp
```

What's Next?

```
C:\Users\Lenovo>docker ps
```

```
C:\Users\Lenovo>docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
1c17503a759c	170ale90f843	"docker-entrypoint.s..."	5 minutes ago	Up 22 seconds	6379/
tcp	amazing_robinson				
a388ed3e2ae7	docker/dev-environments-default:stable-1	"sleep infinity"	23 months ago	Up 26 seconds	
siddhesh-app-1					

```
C:\Users\Lenovo>docker stop 1c17503a759c
1c17503a759c
```

```
C:\Users\Lenovo>docker rm 1c17503a759c
1c17503a759c
```

```
C:\Users\Lenovo>docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
redis	latest	f0acdeb5ced3	14 hours ago	140MB
ubuntu	latest	802541663949	6 months ago	78.1MB
ubuntu	<none>	ca2b0f26964c	24 months ago	77.9MB
redis	<none>	170a1e90f843	2 years ago	138MB
docker/dev-environments-default	stable-1	7eb85f44c229	3 years ago	610MB
crazymax/linguist	7.20.0	dc5f9ae3be6a	3 years ago	72.2MB
docker/desktop-git-helper	5a4fca126aadcd3f6cc3a011aa991de982ae7000	efe2d67c403b	4 years ago	44.2MB

```
C:\Users\Lenovo>docker rmi f0acdeb5ced3
Untagged: redis:latest
Untagged: redis@sha256:dc123d5a424cbe44d681822c2a620f3f44bdba240b8a58a0eee0048a285532ed
Deleted: sha256:f0acdeb5ced3472faa8a41497498730d1b30e52ab4b8bf867b5b5ddb87720ec6
Deleted: sha256:ef393d323f971e457b565382266ff95d29e75cdcb8fa437e9bf7b73fc7313c11
Deleted: sha256:7ca2926f5c45fc35d3d02d66ad386ff0e4067527df8ca12b1b2189b185ddcfbc
Deleted: sha256:365e4cce10e83a6bb3cf41f881e4869ab7cdbaa65371796f090558e1ca2aae7e
Deleted: sha256:42a797d73e006fc25a5ca6dc274c835c061f206832948154875f994fabfb2d50
Deleted: sha256:fba38928ef78663422aca98187e21a3f0012559ec1a73c776b2de370967c45d3
Deleted: sha256:b599de35d6f51c157a9d80d634e4f2f4751301df3d2498f5acb35167a8d34642
Deleted: sha256:a8ff6f8cbdf6741c10dd183560df7212db666db046768b0f05bbc3904515f03
```

```
C:\Users\Lenovo>docker run --name ubuntu --rm -i -t ubuntu bash
root@0561ed697d99:/# exit
exit
```

```
C:\Users\Lenovo>docker run -d -p 8080:8080 ubuntu
f2de7b7b95eddddf5d839545d2e360376f90dee740084e1874c5c2c103bdcf1c
docker: Error response from daemon: Ports are not available: exposing port TCP 0.0.0.0:8080 -> 0.0.0.0:0: listen tcp 0.0.0.0:8080: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.
```

Conclusion : Successfully implemented Dockerfile instructions, build an image for a sample web application using DOCKERFILE